

Siwon Kim

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github | orcid
Seoul, Korea

OBJECTIVE

Seeking a challenging position in Data Science and AI Engineering directed toward practical impact. An engineering researcher who proactively builds through obstacles; an enthusiastic and adaptive learner with a physics/mathematics background, focusing on AI engineering and electrocardiogram analysis. Recent research interests include self-supervised learning and hierarchical representations.

EXPERIENCE

- **Postdoctoral Researcher** Mar 2026 – Present
Research Institute of Basic Sciences, Seoul National University

EDUCATION

- **Seoul National University** March 2021 – Feb 2026
Ph.D in Mathematical Science (Integrated) Advisor: Woong Kook
Seoul, Korea
- **Konkuk University** March 2017 – Feb 2021
B.S, Double major in Physics and Mathematics
Seoul, Korea

PROJECTS

- **AI-Assisted Rule-Based ECG Diagnosis Software with MirabellSoft Inc.** Jan 2024 – Present
AI-assisted ECG delineation GUI tool with interpretable rule-based diagnosis for educational purposes
Main Engineer | Team of 2
 - Improved ECG delineation AI model for complex conduction block arrhythmias
 - Developed an AI-assisted GUI application for labeling ECG delineation data
 - Engineered an ECG processing & feature calculation API for rule-based classification research
 - Implemented, developed, and tested ECG classification rules
- **ECG Arrhythmia Detection with MediFarmSoft Co., Ltd** Jan 2023 – Jan 2024
AI-driven diagnosis of arrhythmia using features extracted from AI ECG delineations
Core Engineer | Team of 5
 - Developed initial AFIB/AFL detection model and subsequently implemented features for a unified arrhythmia classifier
 - Managed version control and maintained source code architecture
 - Authored technical documentation and facilitated direct communication with MediFarmSoft
- **Anomaly Detection on Semiconductor Manufacturing Processes with SimPlatform Inc.** July 2023 – Feb 2024
Research on preventative measures using high-volume, raw time-series sensor data
Assistant Engineer | Team of 3
 - Analyzed a CVD process using moving averages and z-score peak detection to identify anomalies
- **Subtyping of AFIB ECG with SNU College of Medicine** Oct 2023 – May 2024
Classification task between paroxysmal and persistent AFIB with limited data (~700 samples)
Main Engineer | Team of 3
 - Evaluated advanced dynamical system features of ECG using mathematical methods
 - Synthesized team analyses and led communication with the college

PATENTS & PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [S.1] Siwon Kim (2026). Hierarchical Self-Supervised Representation Learning Framework for Multivariate Time Series Grounded in ECG Analysis. Manuscript submitted for publication in *Neural Networks*.
- [T.1] Siwon Kim (2026). Outlier Detection with Information Centrality on Transformer Representation Space. Seoul National University

ACADEMIC EXPERIENCE

Contributed Talks

- **Data Analysis with Information Centrality** June 26, 2026
Young Researcher Session in SCoDA Workshop: Statistics on Complex Data - Seoul 2026
Seoul National University
- **Information Centrality: Spectral Embedding Outlier Detector** March 26, 2025
Mid-term Workshop, Samsung DS Basic Science Program (Industry-Academic Cooperation Program)
Seoul National University

Teaching & Service

- **Teaching Assistant** March 2021 – Feb 2026
Seoul National University

- Calculus (5 semesters), Set Theory, Discrete Mathematics, Topological Combinatorics
- 2025 Manager of digital homework system for Seoul National University calculus classes (UNIME)

- **Mentor & Organizing Assistant of HKUST-SNU SPIA 2025**

Summer Research Program in Industrial and Applied Mathematics 2025

1 July – 15 Aug, 2025

Seoul National University

- Mentored the “Heart data analysis via AI” team on a CXR delineation project without labelled data
- Assisted with overall program organization and student logistics

SKILLS

- **Programming Languages:** Python (proficient), Rust & Typescript & Lua (Novice)
- **Data Science & Machine Learning:** Self-Supervised Learning, JEPA, Vision Transformer, U-Net
- **Domain Expertise:** EKG Analysis
- **DevOps & Version Control:** Git, GitHub, CodeBerg (Branching, Merge Conflict Resolution, Overall Maintenance)
- **Other Tools & Technologies:** GUI development (PyQt)

ADDITIONAL INFORMATION

Languages: Korean (Native), English (Fluent)

Interests: Software Development, Code Editor Customization (Neovim), Backcountry Skiing